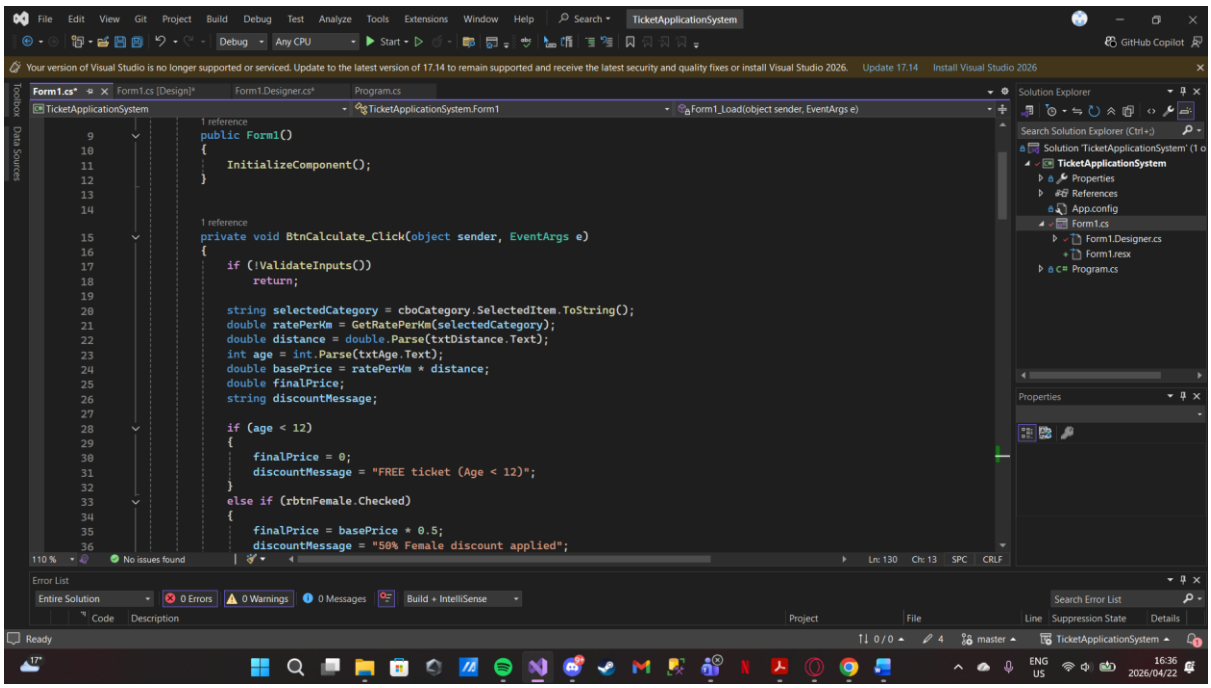
Week 2 Friday project - Ticket Application system

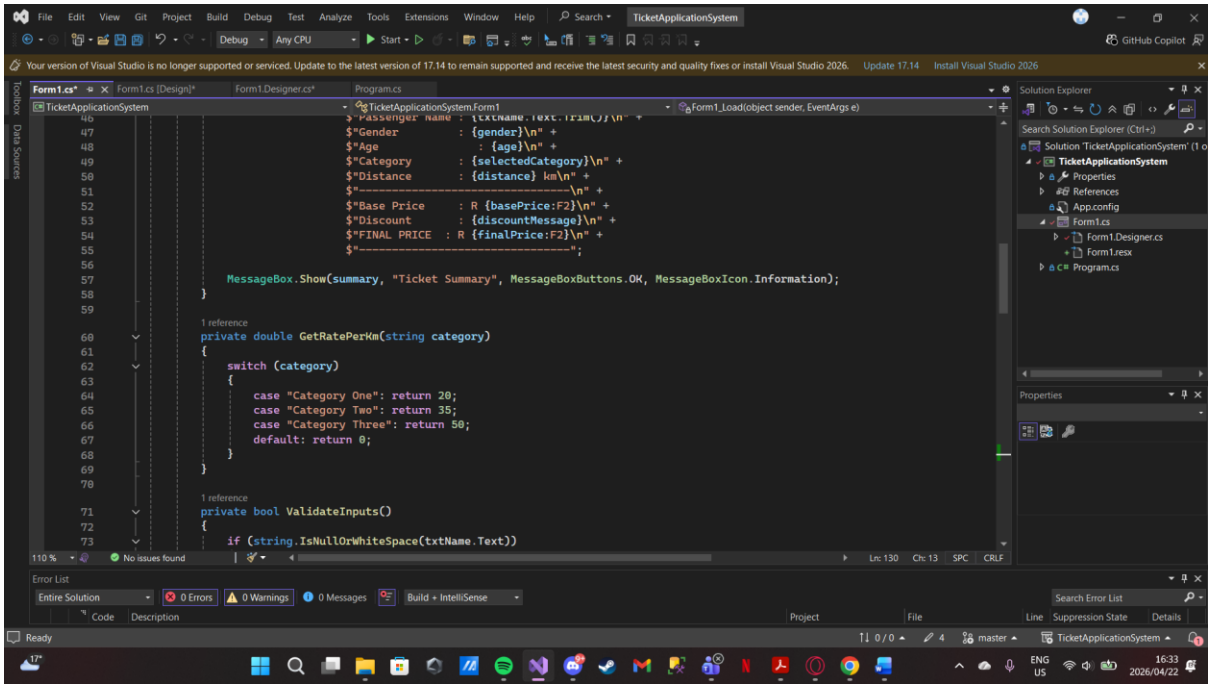
Group Members: Zander du Plessis, Josh Jansen van Rensburg and Stern George.

Screenshots:

Calculations with added discounts:



```
public Form1()
{
    InitializeComponent();
}


private void BtnCalculate_Click(object sender, EventArgs e)
{
    if (!ValidateInputs())
        return;

    string selectedCategory = cboCategory.SelectedItem.ToString();
    double ratePerKm = GetRatePerKm(selectedCategory);
    double distance = double.Parse(txtDistance.Text);
    int age = int.Parse(txtAge.Text);
    double basePrice = ratePerKm * distance;
    double finalPrice;
    string discountMessage;

    if (age < 12)
    {
        finalPrice = 0;
        discountMessage = "FREE ticket (Age < 12)";
    }
    else if (rbtnFemale.Checked)
    {
        finalPrice = basePrice * 0.5;
        discountMessage = "50% Female discount applied";
```



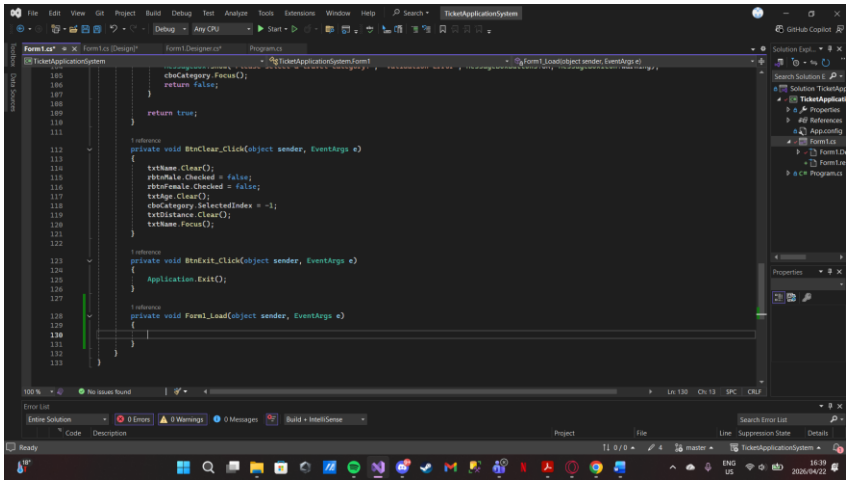
```
$"Gender          : {gender}\n" +
$"Age               : {age}\n" +
$"Category        : {selectedCategory}\n" +
$"Distance        : {distance} km\n" +
$"--------------------------------\n" +
$"Base Price      : R {basePrice:F2}\n" +
$"Discount        : {discountMessage}\n" +
$"FINAL PRICE  : R {finalPrice:F2}\n" +
$"--------------------------------";

    MessageBox.Show(summary, "Ticket Summary", MessageBoxButtons.OK, MessageBoxIcon.Information);
}

private double GetRatePerKm(string category)
{
    switch (category)
    {
        case "Category One": return 20;
        case "Category Two": return 35;
        case "Category Three": return 50;
        default: return 0;
    }
}

private bool ValidateInputs()
{
    if (string.IsNullOrWhiteSpace(txtName.Text))
```

Input validation:

```
                default: return 0;
            }
        }

        1 reference
        private bool ValidateInputs()
        {
            if (string.IsNullOrWhiteSpace(txtName.Text))
            {
                MessageBox.Show("Please enter passenger name.", "Validation Error", MessageBoxButtons.OK, MessageBoxIcon.Warning);
                txtName.Focus();
                return false;
            }

            if (!rbtnMale.Checked && !rbtnFemale.Checked)
            {
                MessageBox.Show("Please select gender (MALE or FEMALE).", "Validation Error", MessageBoxButtons.OK, MessageBoxIcon.Warning);
                return false;
            }

            if (!int.TryParse(txtAge.Text, out int age) || age <= 0 || age > 120)
            {
                MessageBox.Show("Please enter a valid age (1-120).", "Validation Error", MessageBoxButtons.OK, MessageBoxIcon.Warning);
                txtAge.Clear();
                txtAge.Focus();
                return false;
            }

            if (!double.TryParse(txtDistance.Text, out double distance) || distance <= 0)
            {
                MessageBox.Show("Please enter a valid distance (> 0 km).", "Validation Error", MessageBoxButtons.OK, MessageBoxIcon.Warning);
                txtDistance.Clear();
                txtDistance.Focus();
                return false;
```

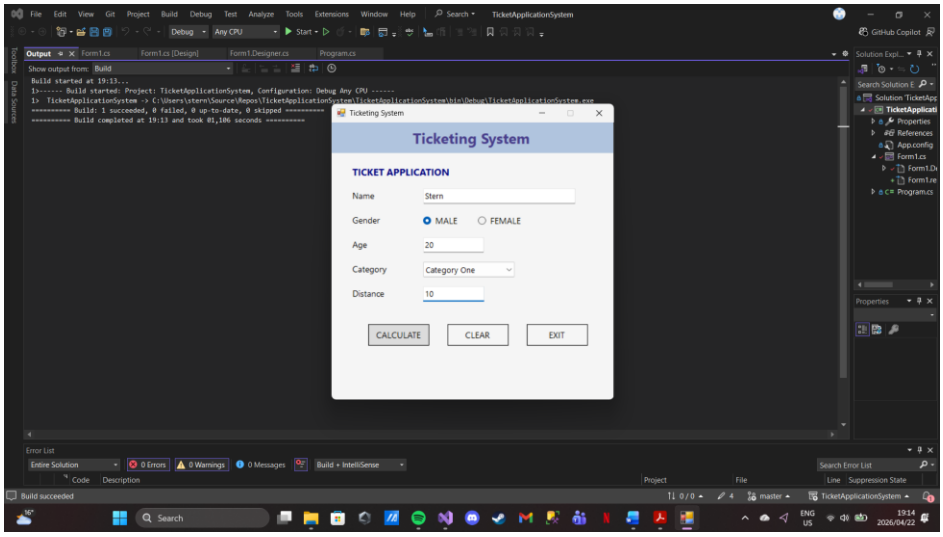
```
            {
                MessageBox.Show("Please select gender (MALE or FEMALE).", "Validation Error", MessageBoxButtons.OK, MessageBoxIcon.Warning);
                return false;
            }

            if (!int.TryParse(txtAge.Text, out int age) || age <= 0 || age > 120)
            {
                MessageBox.Show("Please enter a valid age (1-120).", "Validation Error", MessageBoxButtons.OK, MessageBoxIcon.Warning);
                txtAge.Clear();
                txtAge.Focus();
                return false;
            }

            if (!double.TryParse(txtDistance.Text, out double distance) || distance <= 0)
            {
                MessageBox.Show("Please enter a valid distance (> 0 km).", "Validation Error", MessageBoxButtons.OK, MessageBoxIcon.Warning);
                txtDistance.Clear();
                txtDistance.Focus();
                return false;
            }

            if (cboCategory.SelectedIndex == -1)
            {
                MessageBox.Show("Please select a travel category.", "Validation Error", MessageBoxButtons.OK, MessageBoxIcon.Warning);
                cboCategory.Focus();
                return false;
            }

            return true;
        }

        1 reference
        private void BtnClear_Click(object sender, EventArgs e)
        {
```

Buttons working:

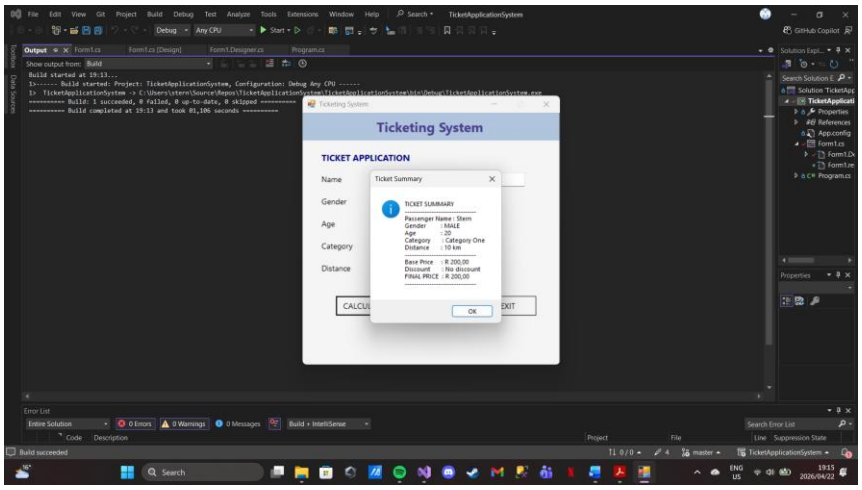


Application screenshots:

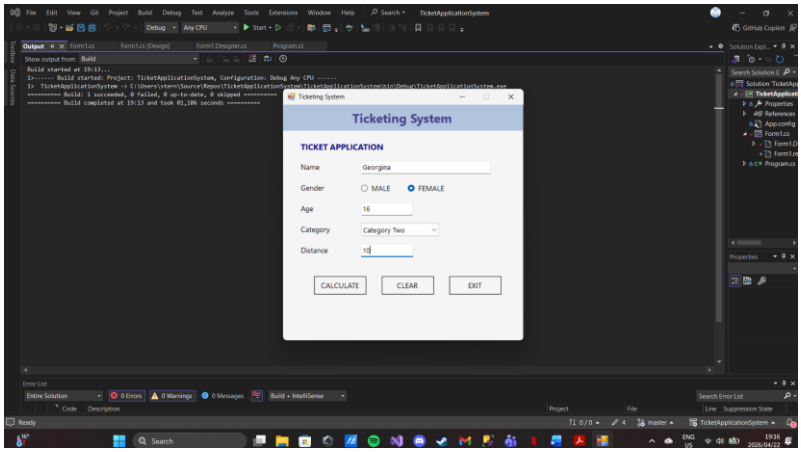
Example 1:



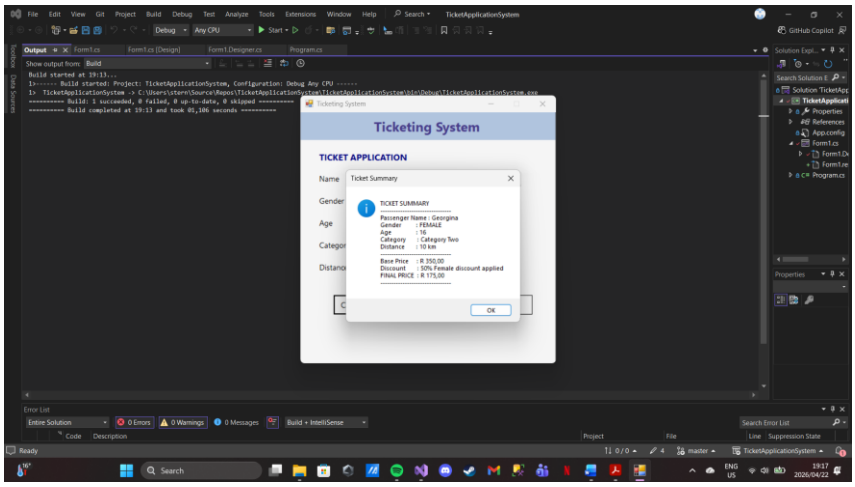
Results:



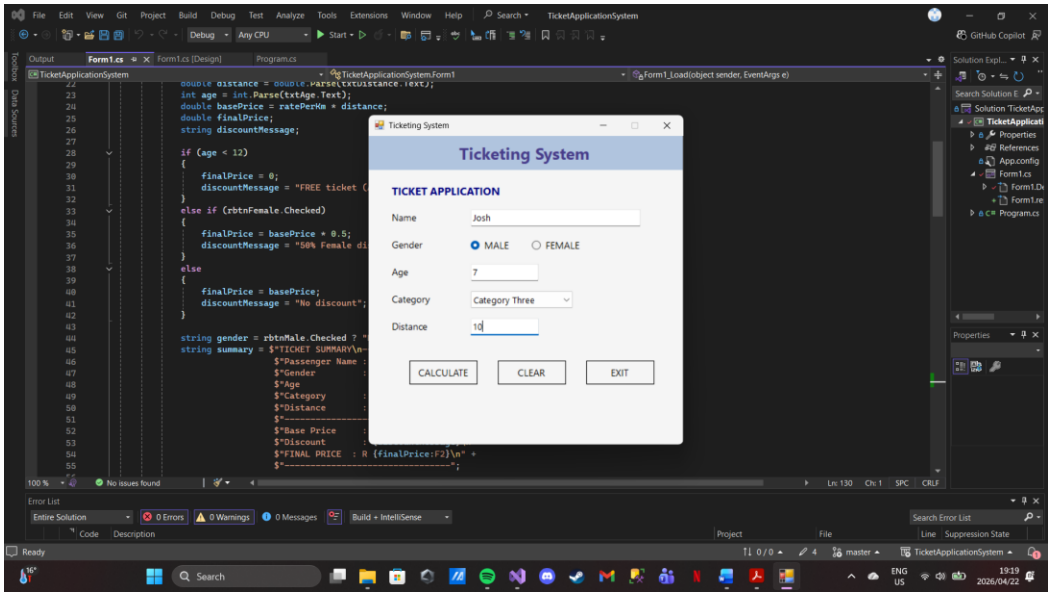
Example 2:



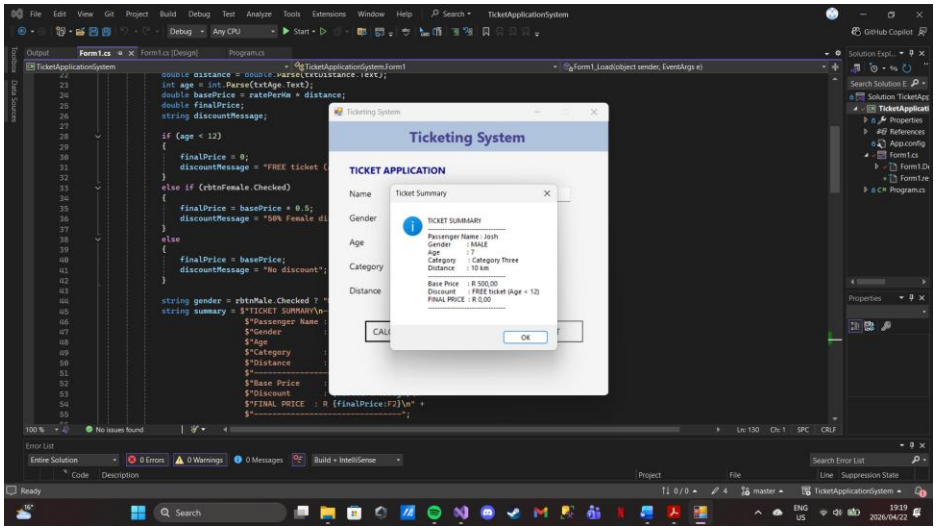
Results:



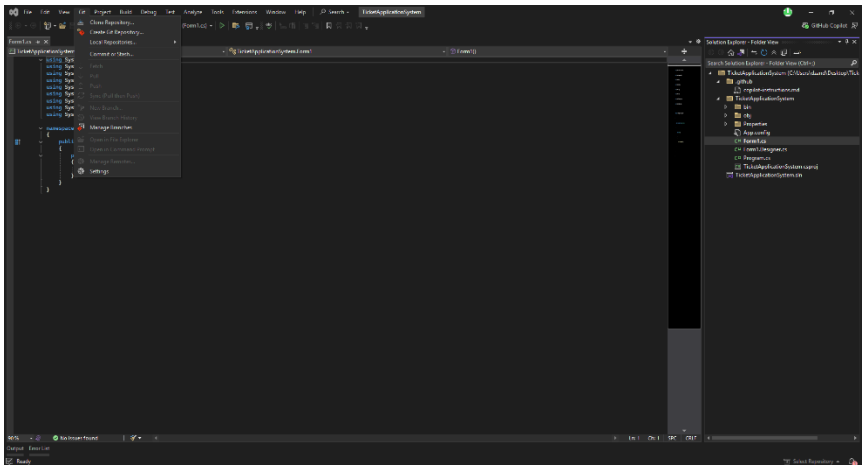
Example 3:

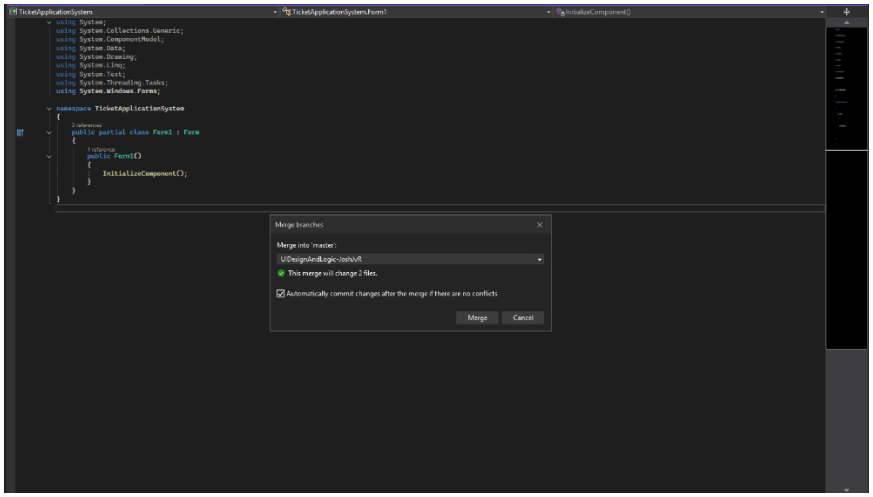


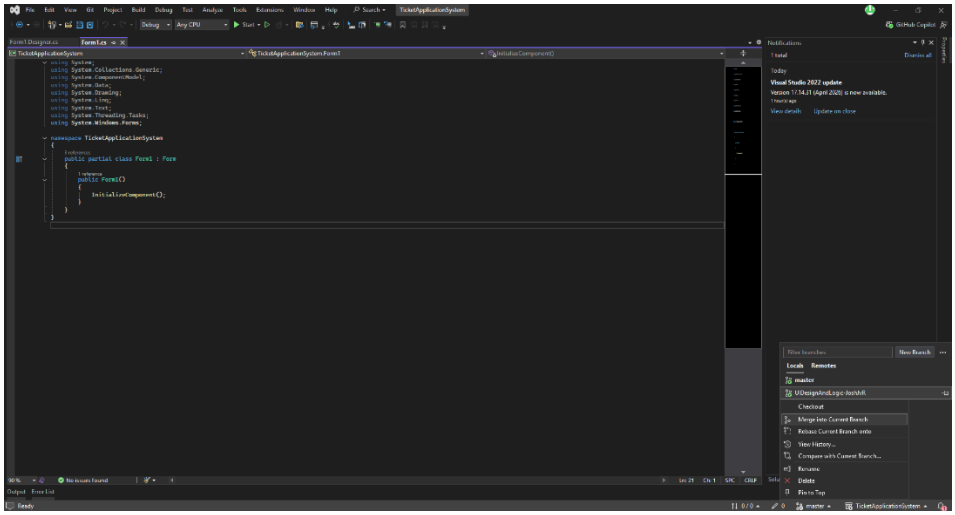
Results:

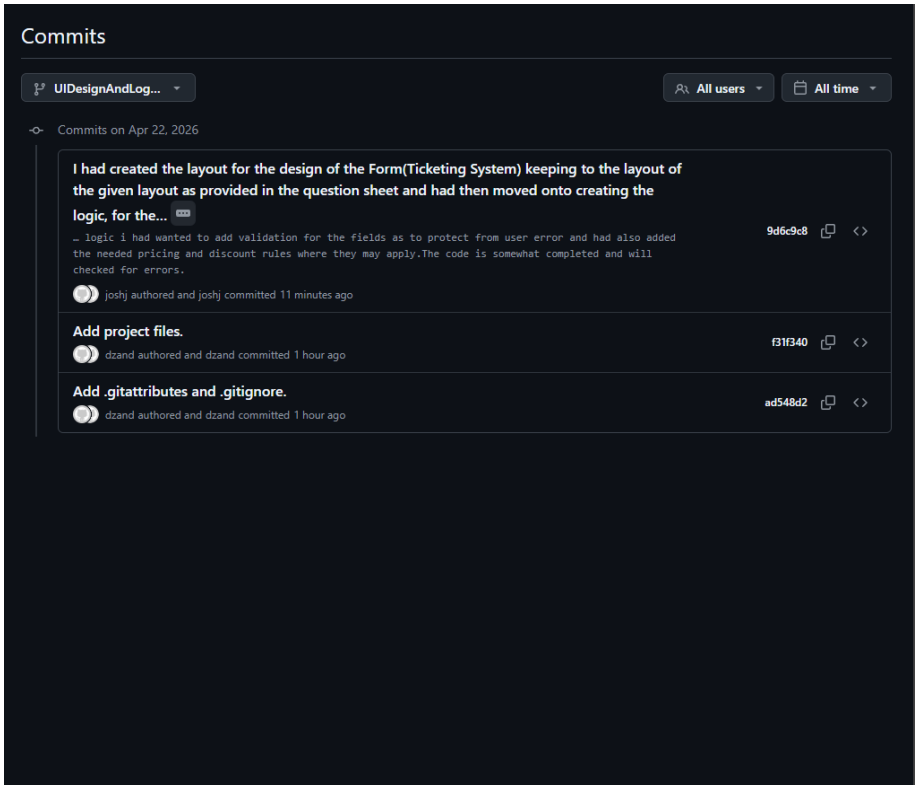


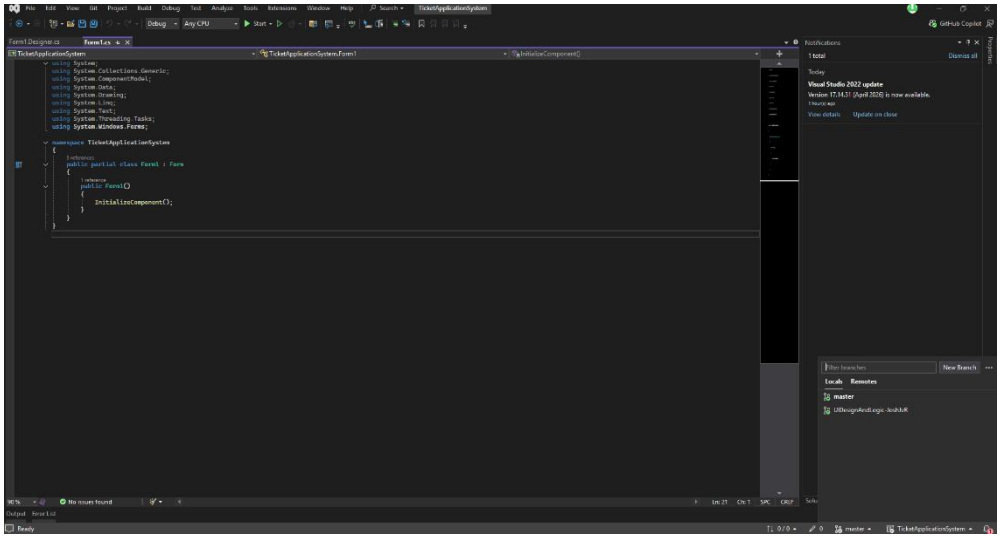
Github setup:

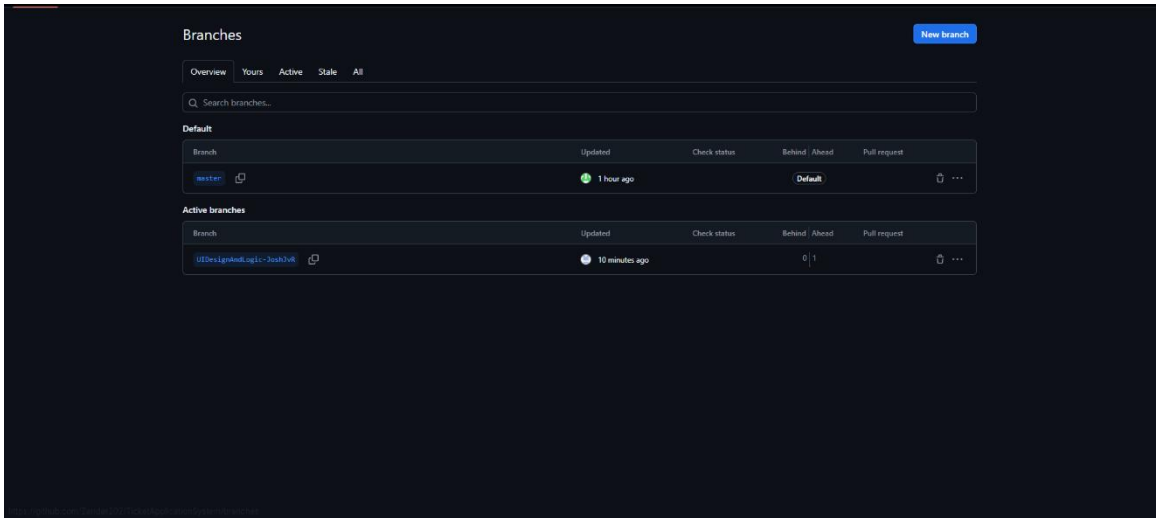


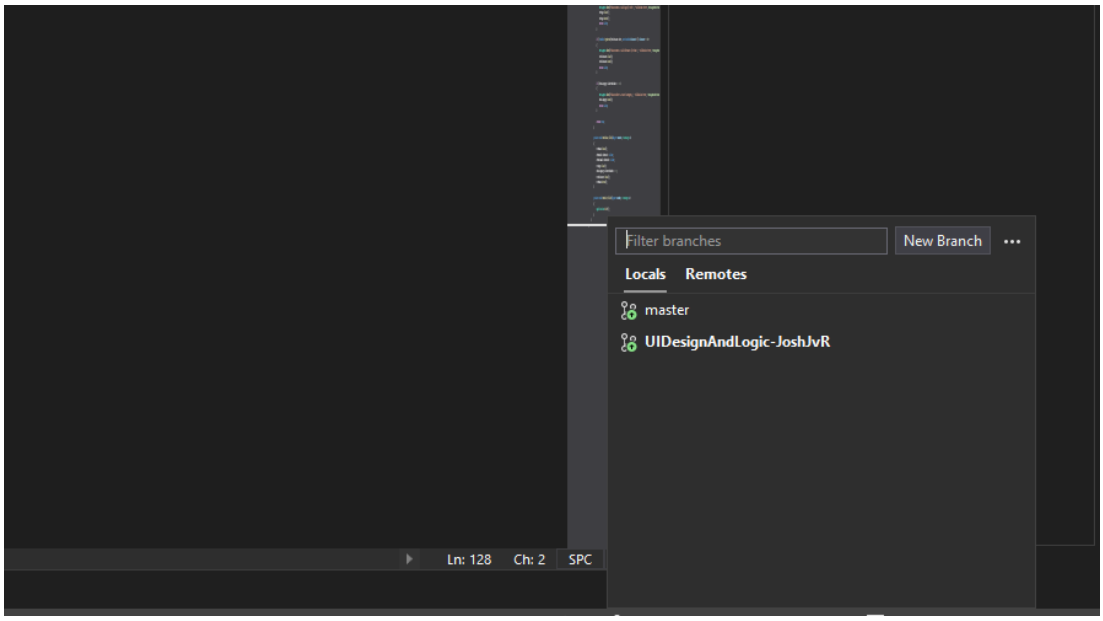


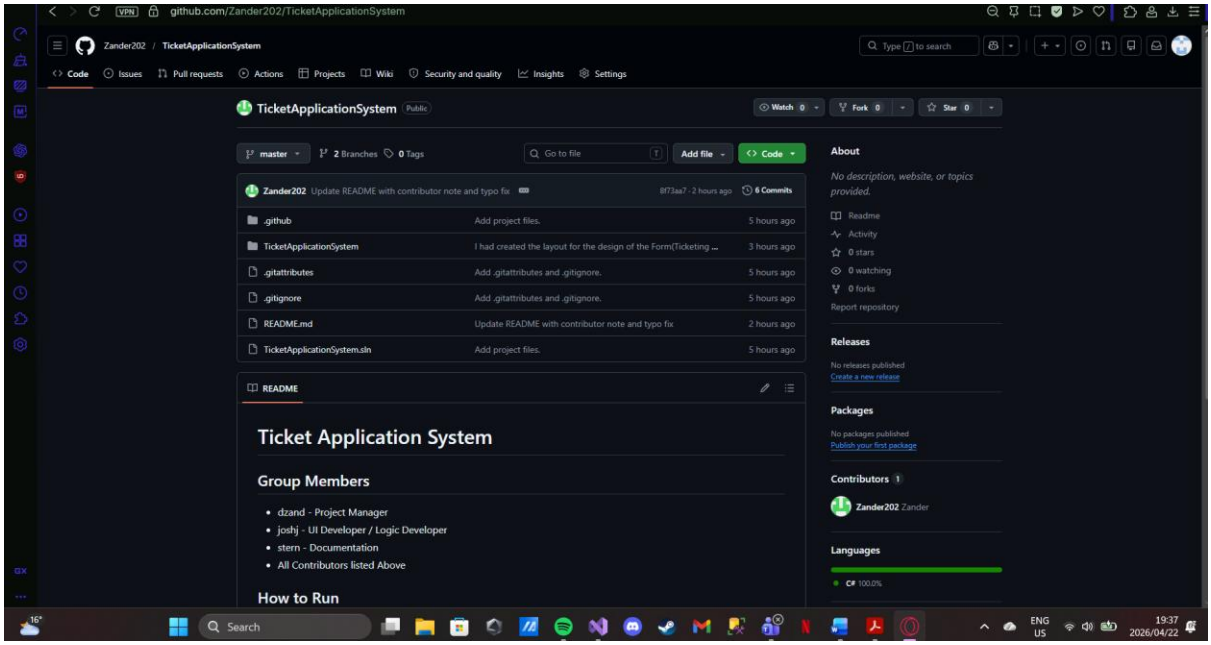

File Edit View Git Project Build Debug Test Analyze Tools Extensions Window Help
TicketApplicationSystem
Debug
Any CPU
Start
Form1.Designer.cs
Form1.cs
TicketApplicationSystem
TicketApplicationSystem.Form1
InitializeComponent()
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
namespace TicketApplicationSystem
{
public partial class Form1 : Form
{
public Form1()
{
InitializeComponent();
}
}
}
Notifications
1 total
Dismiss all
Today
Visual Studio 2022 update
Version 17.14.31 (April 2026) is now available.
1 hour ago
View details
Update on close
Filter branches
New Branch
Local
Remotes
master
UIDesignAndLogic-Joshua
Checkout
Merge into Current Branch
Rebase Current Branch onto
View History...
Compare with Current Branch...
Rename
Delete
Pin to Top
No issues found
Ln: 21
Ch: 1
SPC
CRLF
Output
Error List
Ready
master
TicketApplicationSystem


Commits
UIDesignAndLog...
All users
All time
Commits on Apr 22, 2026
I had created the layout for the design of the Form(Ticketing System) keeping to the layout of the given layout as provided in the question sheet and had then moved onto creating the logic, for the...
… logic i had wanted to add validation for the fields as to protect from user error and had also added the needed pricing and discount rules where they may apply.The code is somewhat completed and will checked for errors.
joshj authored and joshj committed 11 minutes ago
9d6c9c8
Add project files.
dzand authored and dzand committed 1 hour ago
f31f340
Add .gitattributes and .gitignore.
dzand authored and dzand committed 1 hour ago
ad548d2


Form1.Designer.cs
Form1.cs
TicketApplicationSystem
TicketApplicationSystem.Form1
InitializeComponent()
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
namespace TicketApplicationSystem
{
public partial class Form1 : Form
{
public Form1()
{
InitializeComponent();
}
}
}
No issues found
Notifications
1 total
Dismiss all
Today
Visual Studio 2022 update
Version: 17.14.21 (April 2026) is now available.
1 hours ago
View details
Update on close
Filter branches
New Branch
Locals
Remotes
master
UIDesignAndLogic-JoshJvR
master
TicketApplicationSystem


Branches
New branch
Overview
Yours
Active
Stale
All
Search branches...
Default
Branch
Updated
Check status
Behind
Ahead
Pull request
master
1 hour ago
Default
Active branches
Branch
Updated
Check status
Behind
Ahead
Pull request
UIDesignAndLogic-JoshJvR
10 minutes ago
0
1


Filter branches
New Branch
Locals
Remotes
master
UIDesignAndLogic-JoshJvR
Ln: 128
Ch: 2
SPC


github.com/Zander202/TicketApplicationSystem
Zander202 / TicketApplicationSystem
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
TicketApplicationSystem
Public
master
2 Branches
0 Tags
Go to file
Add file
Code
Zander202 Update README with contributor note and typo fix
6 Commits
.github
Add project files.
5 hours ago
TicketApplicationSystem
I had created the layout for the design of the Form(Ticketing ...
3 hours ago
.gitattributes
Add .gitattributes and .gitignore.
5 hours ago
.gitignore
Add .gitattributes and .gitignore.
5 hours ago
README.md
Update README with contributor note and typo fix
2 hours ago
TicketApplicationSystem.sln
Add project files.
5 hours ago
README
Ticket Application System
Group Members
dzand - Project Manager
joshj - UI Developer / Logic Developer
stern - Documentation
All Contributors listed Above
How to Run
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
0 forks
Report repository
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1
Zander202 Zander
Languages
C# 100.0%
Search
ENG
US
19:37
2026/04/22